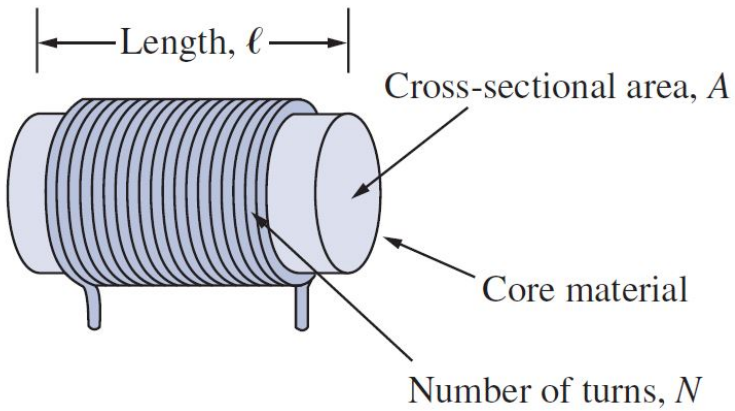
Inductors

- An inductor is a passive element designed to store energy in its magnetic field.
- Inductors find numerous applications in electronic and power systems. They are used in power supplies, transformers, radios, TVs, radars, and electric motors.
- Any conductor of electric current has inductive properties and may be regarded as an inductor. But in order to enhance the inductive effect, a practical inductor is usually formed into a cylindrical coil with many turns of conducting wire, as shown in the figure below.



Inductors

- An inductor consists of a coil of conducting wire.
- If current is allowed to pass through an inductor, it is found that the voltage across the inductor is directly proportional to the time rate of change of the current. Using the passive sign convention,

$$V = L\frac{di}{dt}$$

where L is the constant of proportionality called the **inductance** of the inductor. The unit of inductance is the henry (H), named in honor of the American inventor Joseph Henry (1797–1878).

Inductance

Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

Inductors

- The inductance of an inductor depends on its physical dimension and construction. Formulas for calculating the inductance of inductors of different shapes are derived from electromagnetic theory and can be found in standard electrical engineering handbooks. For instance, for the inductor shown in the previous figure

$$L = \frac{N^2 \mu A}{l}$$

where N is the number of turns, l is the length, A is the cross-sectional area, and μ is the permeability of the core.

- Note that inductance can be increased by increasing the number of turns of coil, using material with higher permeability as the core, increasing the cross-sectional area, or reducing the length of the coil.

Inductors

- Like capacitors, commercially available inductors come in different values and types. Typical practical inductors have inductance values ranging from a few microhenrys (μH), as in communication systems, to tens of henrys (H) as in power systems. Inductors may be fixed or variable. The core may be made of iron, steel, plastic, or air. The terms *coil* and *choke* are also used for inductors. Common inductors are shown in the figure below.

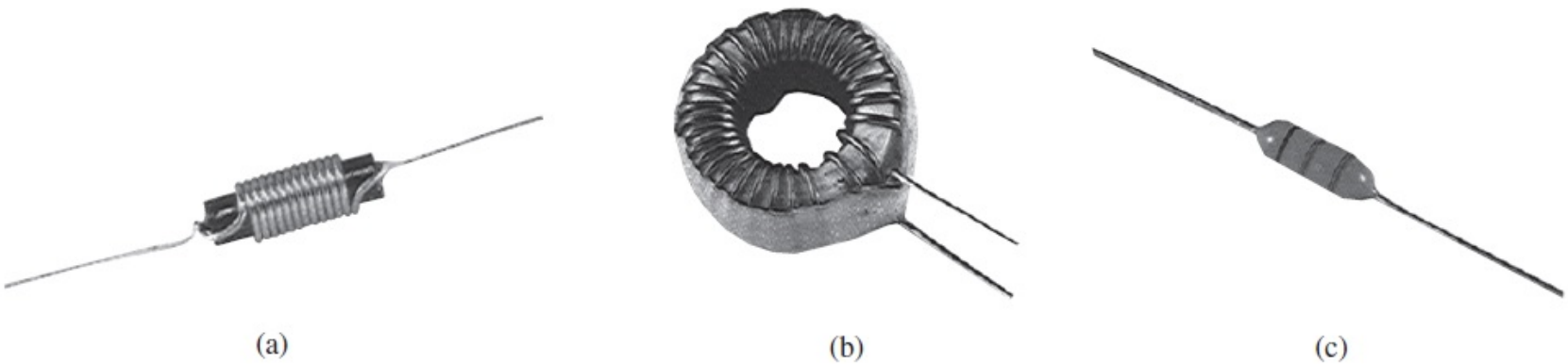


Figure: Various types of inductors: (a) solenoidal wound inductor, (b) toroidal inductor, (c) chip inductor.

Inductors

- The circuit symbols for inductors are shown in the below figure, following the passive sign convention.

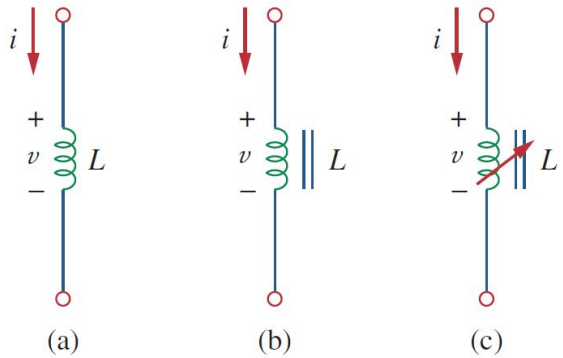
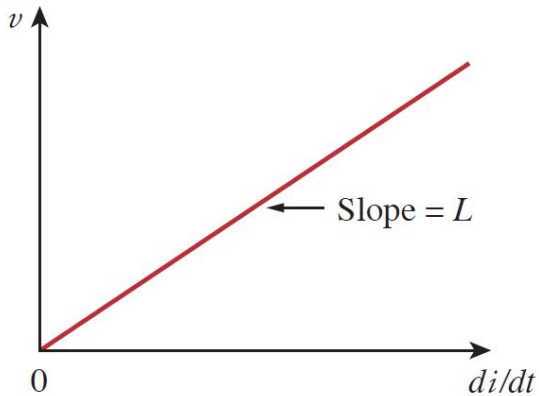


Figure: Circuit symbols for inductors: (a) air-core, (b) iron-core, (c) variable iron-core.

- The voltage-current relationship for an inductor is given in the below figure.



Inductors

- The current-voltage relationship is obtained from $V = Ldi/dt$ as

$$di = \frac{1}{L} v dt$$

Integrating gives

$$i = \frac{1}{L} \int_{-\infty}^{t} v(\tau) d\tau$$

or

$$i = \frac{1}{L} \int_{t_0}^{t} v(\tau) d\tau + i(t_0)$$

where $i(t_0$)is the total current for $-\infty < t < t_0$ and $i(-\infty) = 0$. The idea of making $i(-\infty) = 0$ is practical and reasonable, since there must be a time in the past when there was no current in the inductor.

Inductors

- The inductor is designed to store energy in its magnetic field. The energy stored can be obtained from $v = Ldi/dt$. The power delivered to the inductor is

$$p = vi = \left(L\frac{di}{dt}\right) i$$

The energy stored is

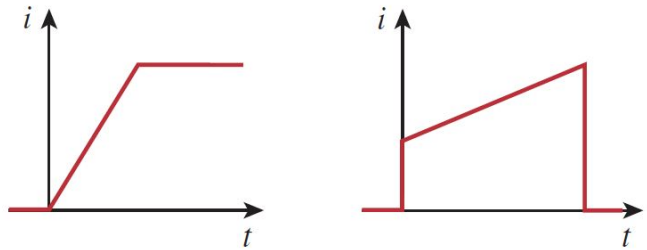
$$w = \int_{-\infty}^{t} p(\tau)d\tau = L\int_{-\infty}^{t} \frac{di}{d\tau} i d\tau$$
$$= L\int_{-\infty}^{t} i di = \frac{1}{2}Li^2(t) - \frac{1}{2}Li^2(-\infty)$$

Since $i(-\infty) = 0$,

$$w = \frac{1}{2}Li^2$$

Inductors

- The important properties of an inductor are as follows:
  1. The voltage across an inductor is zero when the current is constant. Therefore **An inductor acts like a short circuit to DC.**
  2. An important property of the inductor is its opposition to the change in current flowing through it. **The current through an inductor cannot change instantaneously.** For instance, the current through an inductor may take the form shown in figure (a), whereas the inductor current cannot take the form shown in figure (b) in real-life situations due to the discontinuities. However, the voltage across an inductor can change abruptly.



(a) (b)

  3. Like the ideal capacitor, the ideal inductor does not dissipate energy. The energy stored in it can be retrieved at a later time. The inductor takes power from the circuit when storing energy and delivers power to the circuit when returning previously stored energy.

Examples

Example

The current through a $0.1H$ inductor is $i(t) = 10te^{-5t}A$. Find the voltage across the inductor and the energy stored in it.

Examples

Example

The current through a $0.1H$ inductor is $i(t) = 10te^{-5t}A$. Find the voltage across the inductor and the energy stored in it.

Solution:

Since $v = Ldi/dt$ and $L = 0.1H$,

$$v = 0.1\frac{d}{dt}(10te^{-5t}) = e^{-5t} + t(-5)e^{-5t} = e^{-5t}(1 - 5t)V$$

The energy stored is

$$w = \frac{1}{2}Li^2 = \frac{1}{2}(0.1)100t^2e^{-10t} = 5t^2e^{-10t}J$$

Examples

Example

Find the current through a $5H$ inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

Also, find the energy stored at $t = 5s$. Assume $i \cdot v > 0$.

Examples

Example

Find the current through a $5H$ inductor if the voltage across it is

$$v(t) = \begin{cases} 30t^2, & t > 0 \\ 0, & t \leq 0 \end{cases}$$

Also, find the energy stored at $t = 5s$. Assume $i \cdot v > 0$.

Solution:

Since $i = \frac{1}{L}\int_{t_0}^{t} v(t)dt + i(t_0)$ and $L = 5H$,

$$i = \frac{1}{5}\int_{0}^{t} 30t^2 dt + 0 = 6 \times \frac{t^3}{3} = 2t^3 A$$

The power $p = vi = 60t^5$, and the energy stored is then

$$w = \int pdt = \int_{0}^{5} 60t^5 dt = 60\frac{t^6}{6}\Big|_0^5 = 156.25kJ$$

Examples

Solution:
Alternatively, we can obtain the energy stored from

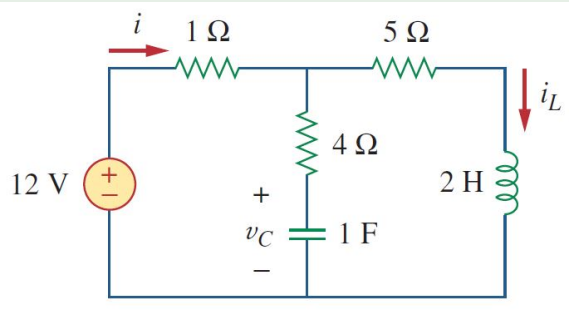
$$w|_0^5 = \frac{1}{2}Li^2(5) - \frac{1}{2}Li(0) = \frac{1}{2}(5)(2 \times 5^3)^2 - 0 = 156.25kJ$$

as obtained before.

Examples

Example

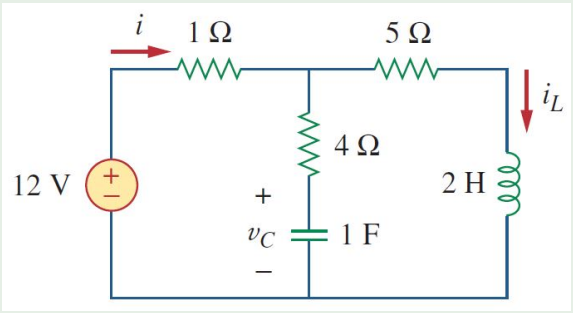
Consider the circuit in the figure below. Under DC conditions, find: (a) i, v_C and i_L (b) the energy stored in the capacitor and inductor.



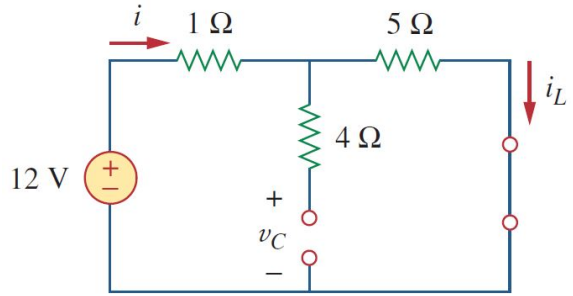
Examples

Example

Consider the circuit in the figure below. Under DC conditions, find: (a) i, v_C and i_L (b) the energy stored in the capacitor and inductor.



Solution: Under DC conditions, we replace the capacitor with an open circuit and the inductor with a short circuit:



Examples

Solution:

- It is evident from the figure that

$$i = i_L = \frac{12}{1+5} = 2A$$

The voltage v_C is the same as the voltage across the 5Ω resistor. Hence,

$$v_C = 5i = 10V.$$

- The energy in the capacitor is

$$w_C = \frac{1}{2}Cv_C^2 = \frac{1}{2}(1)(10^2) = 50J$$

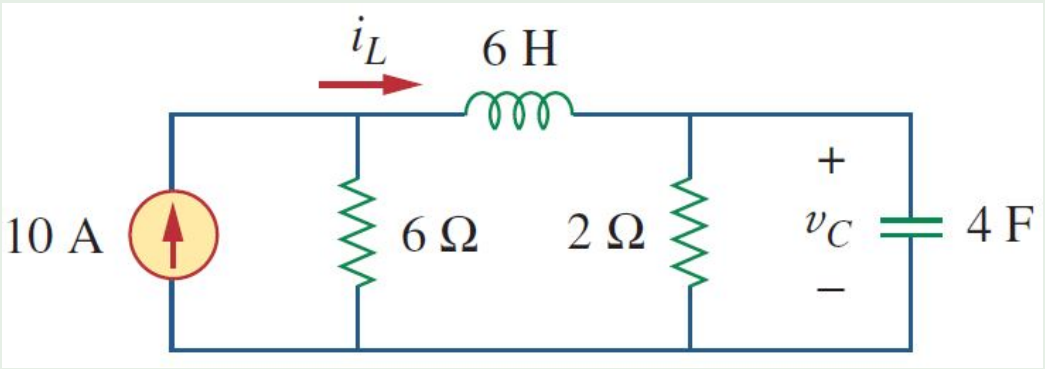
and that in the inductor is

$$w_L = \frac{1}{2}Li_L^2 = \frac{1}{2}(2)(2^2) = 4J$$

Examples

Example

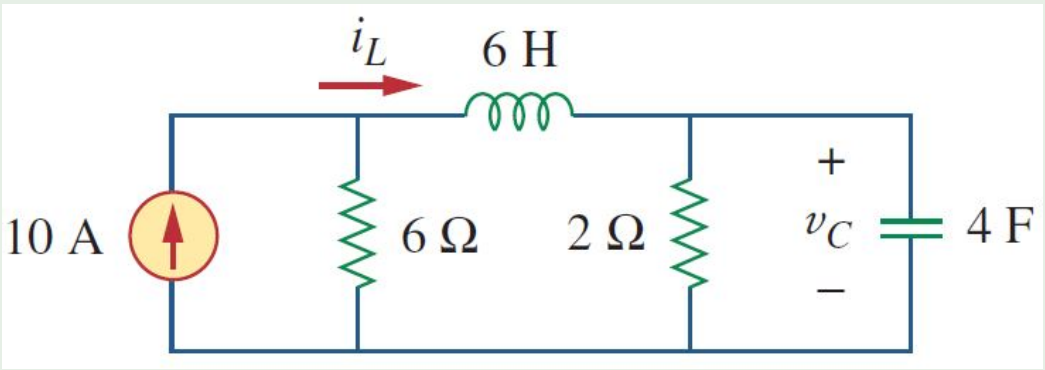
Determine v_C, i_L, and the energy stored in the capacitor and the inductor in the circuit below under DC conditions.



Examples

Example

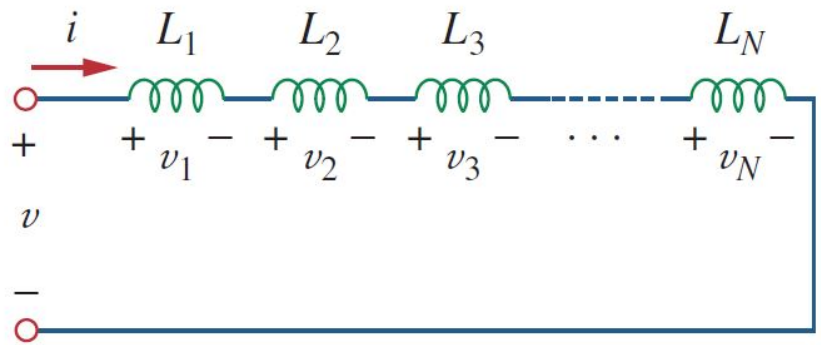
Determine v_C, i_L, and the energy stored in the capacitor and the inductor in the circuit below under DC conditions.



Answer: $v_C = 15V$, $i_L = 7.5A$, $w_C = 450J$ and $w_L = 168.75J$.

Series and Parallel Inductors

- Now that the inductor has been added to our list of passive elements, it is necessary to extend the powerful tool of series-parallel combination.
- We need to know how to find the equivalent inductance of a series-connected or parallel-connected set of inductors found in practical circuits.
- Consider a series connection of N inductors, as depicted in the figure below.



Series and Parallel Inductors

- Applying KVL to the loop, we have

$$v = v_1 + v_2 + \ldots + v_N$$

Substituting $v_k = L_k di/dt$ results in

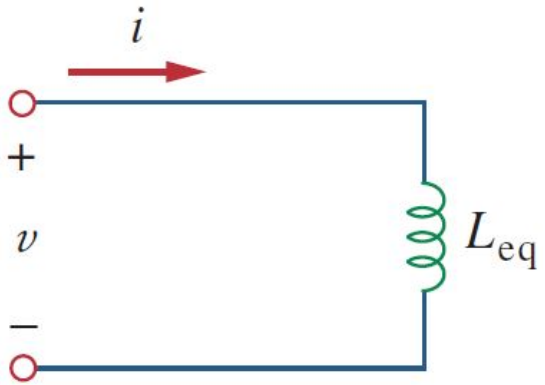
$$\begin{aligned} v &= L_1 \frac{di}{dt} + \ldots + L_N \frac{di}{dt} = (L_1 + \ldots + L_N) \frac{di}{dt} \\ &= \left(\sum_{k=1}^{N} L_k \right) \frac{di}{dt} = L_{eq} \frac{di}{dt} \end{aligned}$$

where

$$L_{eq} = L_1 + \ldots + L_N$$

Series and Parallel Inductors

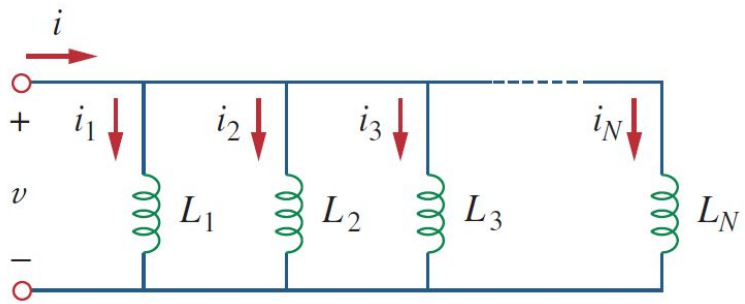
- The equivalent circuit is as shown in the figure below



- The equivalent inductance of series-connected inductors is the sum of the individual inductances.
- Inductors in series are combined in exactly the same way as resistors in series.

Series and Parallel Inductors

- We now consider a parallel connection of N inductors, as shown in the figure below.



- The inductors have the same voltage across them. Using KCL we have

$$i = i_1 + i_2 + \ldots + i_N$$

Series and Parallel Inductors

- However $i_k = \frac{1}{L_k}\int_{t_0}^{t} v dt + i_k(t_0)$. Therefore

$$
\begin{aligned}
i &= \frac{1}{L_1}\int_{t_0}^{t} v dt + i_1(t_0) + \ldots + \frac{1}{L_N}\int_{t_0}^{t} v dt + i_N(t_0) \\
&= \left(\frac{1}{L_1} + \ldots + \frac{1}{L_N}\right)\int_{t_0}^{t} v dt + i_1(t_0) + \ldots + i_N(t_0) \\
&= \left(\sum_{k=1}^{N}\frac{1}{L_k}\right)\int_{t_0}^{t} v dt + \sum_{k=1}^{N} i_k(t_0) = \frac{1}{L_{eq}}\int_{t_0}^{t} v dt + i(t_0)
\end{aligned}
$$

where

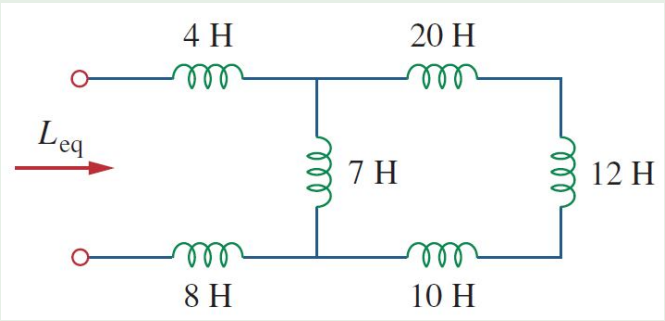
$$
\frac{1}{L_{eq}} = \frac{1}{L_1} + \ldots + \frac{1}{L_N}
$$

- The initial current $i(t_0)$ through L_{eq} at $t = t_0$ is expected by KCL to be the sum of the inductor currents at t_0. Therefore $i(t_0) = i_1(t_0) + \ldots + i_N(t_0)$
- **The equivalent inductance of parallel inductors is the reciprocal of the sum of the reciprocals of the individual inductances.**

Examples

Example

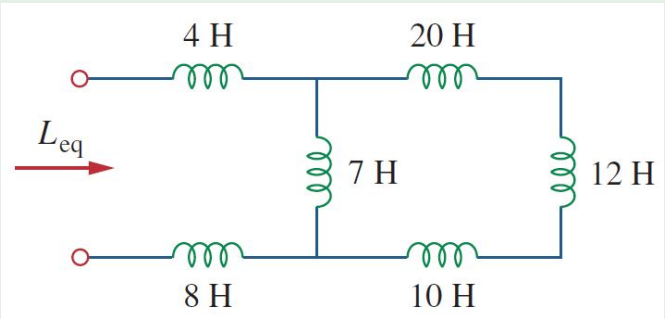
Find the equivalent inductance of the circuit shown in the figure below.



Examples

Example

Find the equivalent inductance of the circuit shown in the figure below.



Solution: The $10H$, $12H$, and $20H$ inductors are in series; thus, combining them gives a $42H$ inductance. This $42H$ inductor is in parallel with the $7H$ inductor so that they are combined, to give

$$\frac{7 \times 42}{7 + 42} = 6H$$

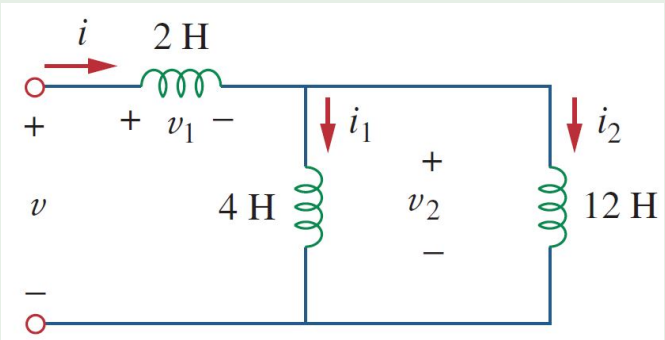
This $6H$ inductor is in series with the $4H$ and $8H$ inductors. Therefore

$$L_{eq} = 4 + 6 + 8 = 18H$$

Examples

Example

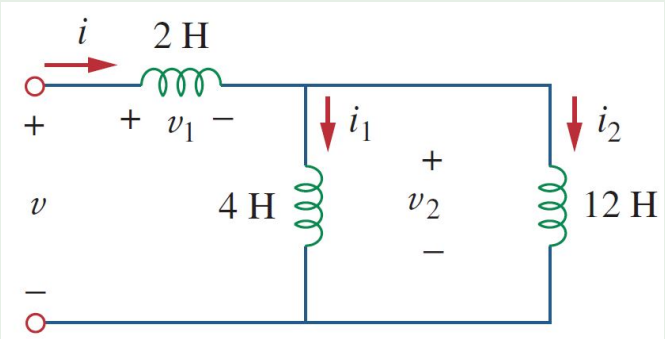
For the circuit in the figure below, $i(t) = 4(2 - e^{-10t})mA$. If $i_2(0) = -1mA$, find a) $i_1(0)$ b) $v(t), v_1(t)$ and $v_2(t)$; c) $i_1(t)$ and $i_2(t)$.



Examples

Example

For the circuit in the figure below, $i(t) = 4(2 - e^{-10t})mA$. If $i_2(0) = -1mA$, find a) $i_1(0)$ b) $v(t), v_1(t)$ and $v_2(t)$; c) $i_1(t)$ and $i_2(t)$.



Solution:

a) From $i(t) = 4(2 - e^{-10t})mA$, we have $i(0) = 4mA$. Since $i = i_1 + i_2$,

$$i_1(0) = i(0) - i_2(0) = 4 - (-1) = 5mA$$

Examples

b) The equivalent inductance is

$$L_{eq} = 2 + (4||12) = 2 + 3 = 5H$$

Therefore

$$v(t) = L_{eq}\frac{di}{dt} = 5(4)(-1)(-10)e^{-10t}mV = 200e^{-10t}mV$$

and

$$v_1(t) = 2\frac{di}{dt} = 2(-4)(-10)e^{-10t}mV = 80e^{-10t}mV$$

Since $v = v_1 + v_2$, we have $v_2(t) = v(t) - v_1(t) = 120e^{-10t}mV$

Examples

c) The current i_1 is obtained as

$$i_1(t) = \frac{1}{4}\int_0^t v_2 dt + i_1(0) = \frac{120}{4}\int_0^t e^{-10t}dt + 5mA$$
$$= -3e^{-10t}|_0^t + 5mA = -3e^{-10t} + 3 + 5 = 8 - 3e^{-10t}mA$$

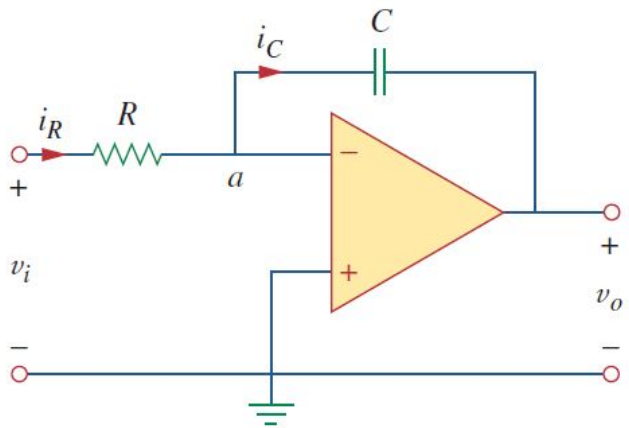
Similarly

$$i_2(t) = \frac{1}{2}\int_0^t v_2 dt + i_2(0) = \frac{120}{12}\int_0^t e^{-10t}dt - 1mA$$
$$= -e^{-10t}|_0^t - 1mA = -e^{-10t} + 1 - 1 = -e^{-10t}mA$$

Note that $i_1(t) + i_2(t) = i(t)$.

Applications of Capacitors and Inductors: Integrator

- Important OPAMP circuits that use energy-storage elements include integrators and differentiators.
- These OPAMP circuits often involve resistors and capacitors; inductors (coils) tend to be more bulky and expensive.
- The OPAMP integrator is used in numerous applications, especially in analog computers.
- Consider the circuit illustrated in the figure below.



Applications of Capacitors and Inductors: Integrator

- Note that the given circuit is similar to inverting amplifier where the feedback resistor is replaced by a capacitor.
- Since input currents of an ideal OPAMP is zero, we have $i_R = i_C$. However,

$$i_R = \frac{v_i}{R}, \quad i_C = -C\frac{dv_o}{dt}$$

Substituting these in $i_R = iC$, we get

$$\frac{v_i}{R} = -C\frac{dv_o}{dt}$$
$$dv_o = -\frac{1}{RC}v_i dt$$

Integrating both sides gives

$$v_o(t) - v_o(0) = -\frac{1}{RC}\int_0^t v_i(\tau)d\tau$$

Applications of Capacitors and Inductors: Integrator

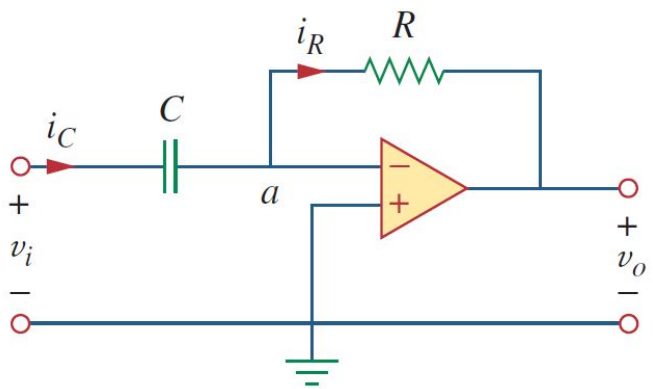
- To ensure that $v_o(0) = 0$, it is always necessary to discharge the integrator's capacitor prior to the application of a signal. Assuming $v_o(0) = 0$, we have

$$v_o = -\frac{1}{RC}\int_0^t v_i(\tau)d\tau$$

  which shows that the circuit provides an output voltage proportional to the integral of the input.
- In practice, the OPAMP integrator requires a feedback resistor to reduce DC gain and prevent saturation. Care must be taken that the OPAMP operates within the linear range so that it does not saturate.

Applications of Capacitors and Inductors: Differentiator

- A differentiator is an OPAMP circuit whose output is proportional to the rate of change of the input signal.
- Consider the circuit illustrated in the figure below.



- Applying KCL at node a, we have $i_R = i_C$.

Applications of Capacitors and Inductors: Differentiator

- However,

$$i_R = -\frac{v_o}{R}, \quad i_C = C\frac{dv_i}{dt}$$

Substituting these in $i_R = i_C$ yields

$$v_o = -RC\frac{dv_i}{dt}$$

showing that the output is the derivative of the input.

- Differentiator circuits are electronically unstable because any electrical noise within the circuit is exaggerated by the differentiator. For this reason, the differentiator circuit discussed in this lecture is not as useful and popular as the integrator. It is seldom used in practice.